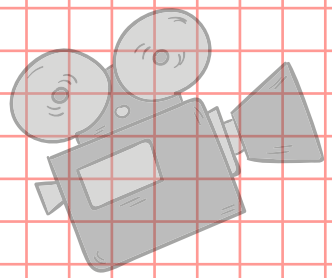
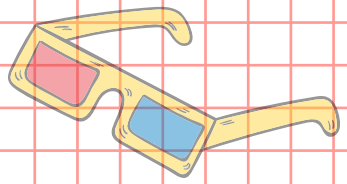
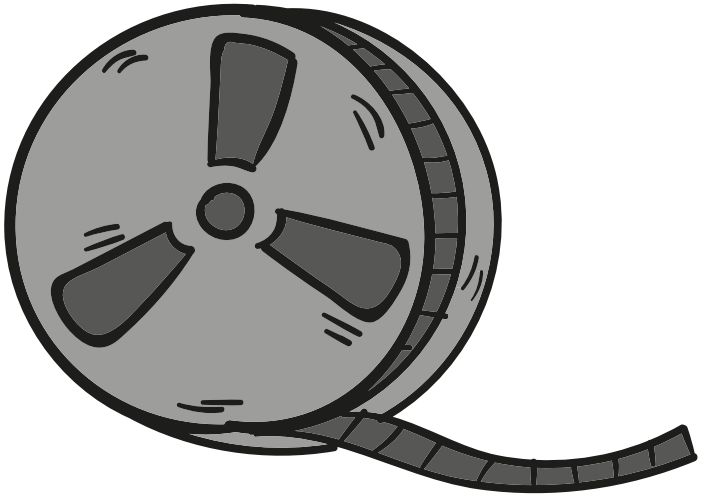
NETFLIX





Netflix Content Strategy & Catalogue Analysis using Python



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01. Project Overview

This project analyses Netflix's content catalogue using Python to understand how the platform has evolved over time and whether its apparent variety reflects genuine diversity or concentration around a smaller set of content patterns.

Rather than stopping at basic descriptive questions, the analysis examines catalogue growth, the balance between movies and TV shows, geographic representation, category concentration, contributor recurrence, and text based description themes. The aim is to move beyond simple charting and develop a more thoughtful view of how Netflix's catalogue appears to be structured and presented.

This project was developed as a portfolio case study to demonstrate practical skills in data cleaning, exploratory data analysis, feature engineering, interpretation, and insight communication.

02. Objectives

01. Understand the structure, quality, and limitations of the Netflix dataset

02. Analyse how the Netflix catalogue has changed over time

03. Examine the balance between movies and TV shows across the catalogue

04. Identify concentration patterns across countries, directors, and cast

05. Use category and text analysis to uncover recurring themes and content patterns

06. Interpret what these combined patterns may suggest about Netflix's content strategy

03. Dataset Overview

The dataset contains **8,807** Netflix titles across 12 variables, with each row representing one content item.

Initial observations:

- The dataset is large enough to support meaningful exploratory analysis
- Several core fields are fully complete, including **type, title, release_year, listed_in,** and **description**
- Missing values exist in director, cast, and country
- Some columns contain multiple values in a single cell, especially **country, cast,** and **listed_in**
- **date_added** and **release_year** represent different concepts and needed to be analysed separately

04. Data Cleaning

To make the dataset analysis ready, several preparation steps were carried out in Python.

Cleaning and feature preparation included:

1. Converting **date_added** from text to datetime format

2. Extracting **year_added, month_added,** and **month_name_added**

3. Splitting **duration** into numeric value and unit

4. Creating separate **duration** fields for movie runtime and TV seasons

5. Splitting and expanding multi value fields such as: **listed_in, country, director, cast**

6. cleaning **title and description text** for later keyword analysis

05.
Analysis & Findings

5.1 Catalogue Evolution Over Time

The first stage of analysis examined how Netflix's catalogue expanded over time using both date_added and release_year.

Key findings:

- Catalogue additions remained small in the early years of the dataset
- Growth accelerated sharply from 2016 onward
- Titles added rose from 418 in 2016 to 1,999 in 2019
- Counts in 2021 are lower, but this likely reflects partial year coverage rather than a confirmed decline

The gap between original release year and Netflix addition year also showed that most titles were added relatively soon after release.

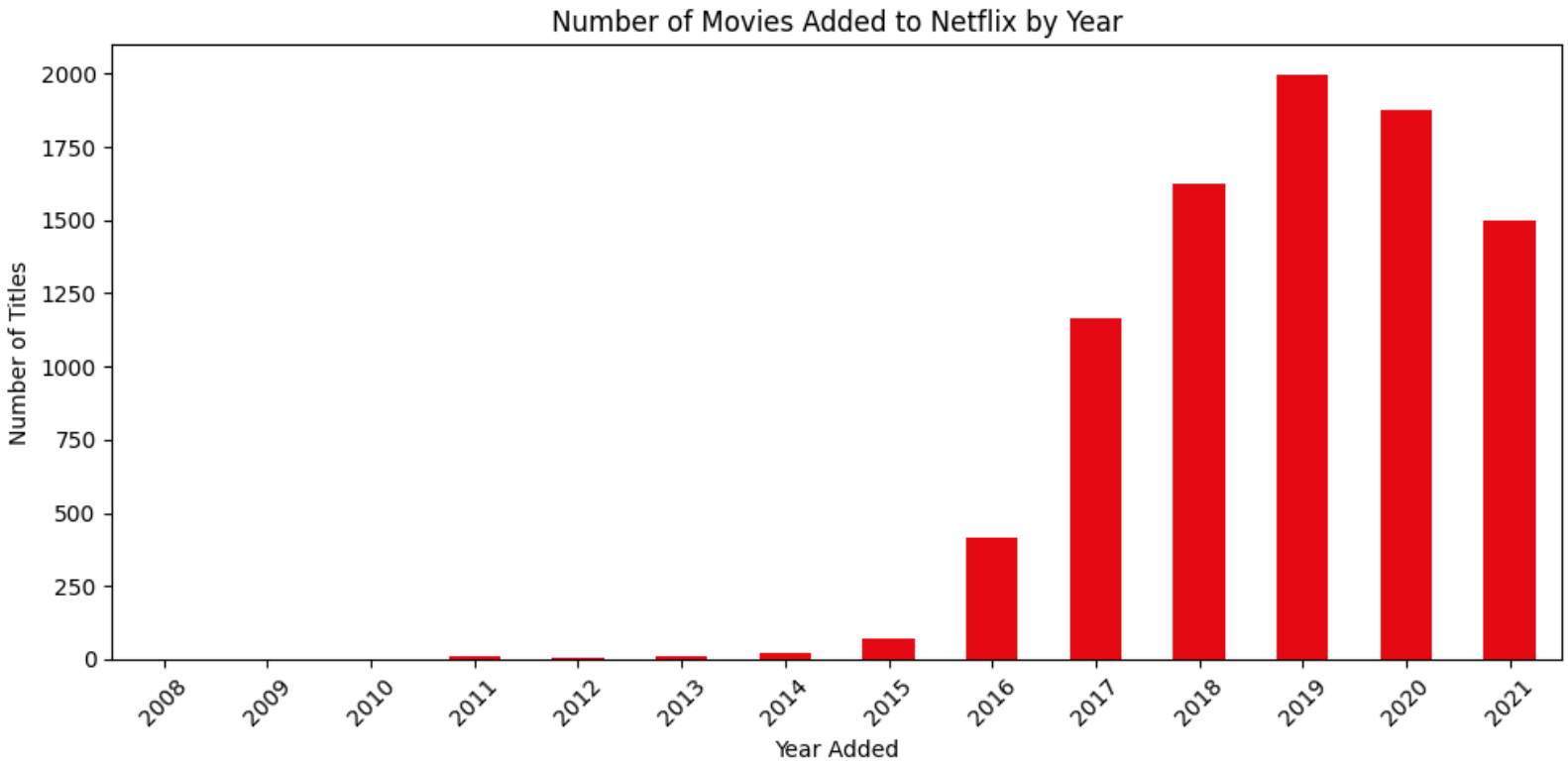
Release to addition timing:

- median gap: 1 year
- mean gap: 4.69 years

This suggests that the catalogue is centred on relatively recent content, while still including a longer tail of older titles added later.

Interpretation:

Netflix's catalogue appears to have gone through a major expansion phase in the late 2010s. Although the library includes older content, the platform seems to rely heavily on relatively recent additions rather than being dominated by historic back catalogue titles.



5.2 Movies vs TV Shows

Overall content mix:

- Movies: 6,131
- TV Shows: 2,676

Key findings:

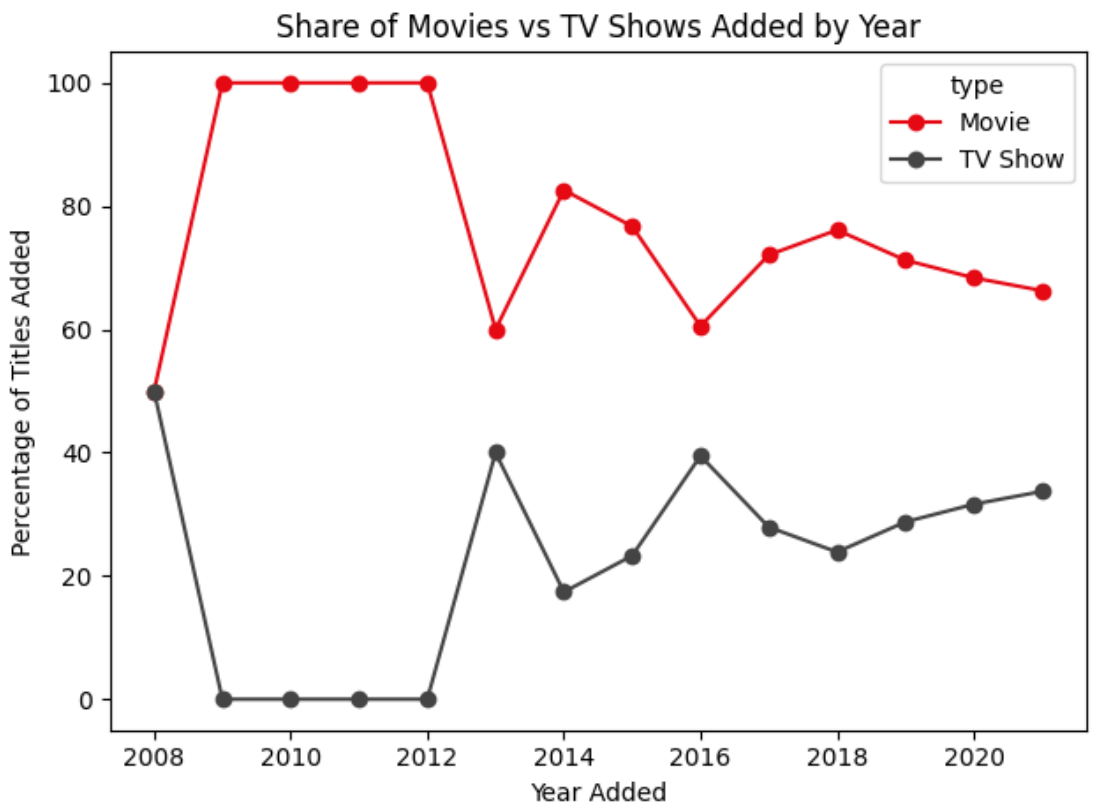
- Movies dominate the catalogue overall
- Movies remained the larger share of annual additions throughout the main growth period
- However, TV shows gained share over time

Share of additions:

- 2018: 76.12% movies, 23.88% TV shows
- 2021: 66.29% movies, 33.71% TV shows

Interpretation:

Netflix remained movie heavy overall, but annual additions became more balanced in later years. This suggests that while films continued to anchor the catalogue, TV content became a more important part of platform growth over time.



5.3 Geographic Concentration and Country Representation

The third stage explored whether Netflix's international catalogue was broadly distributed or concentrated among a smaller set of countries.

Top recorded country participation:

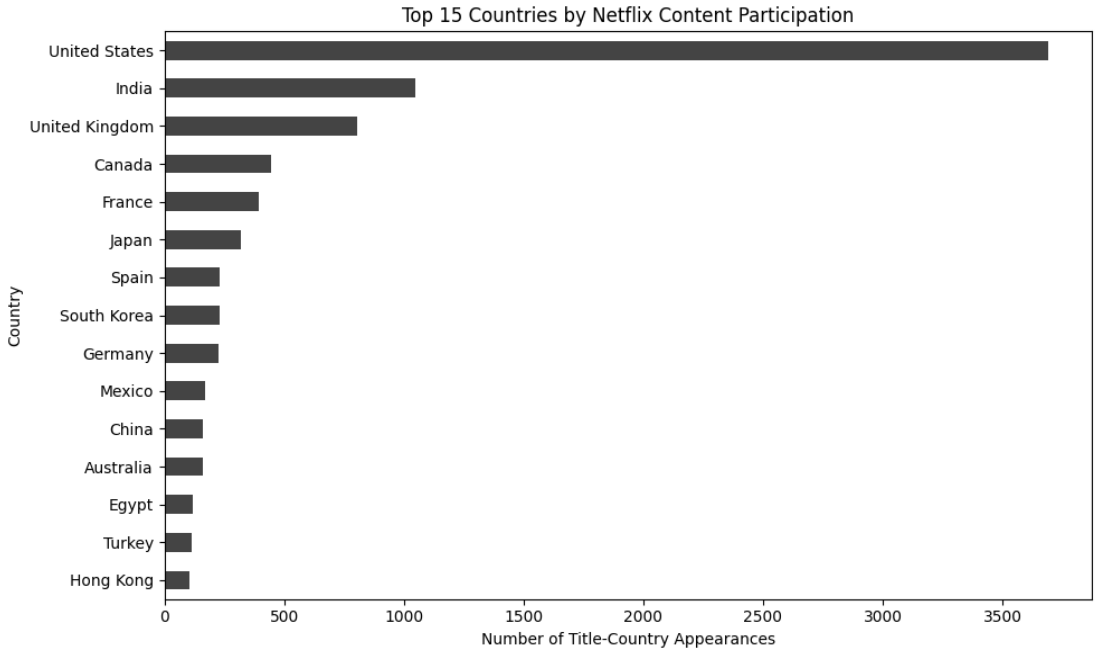
- United States: 3,689
- India: 1,046
- United Kingdom: 804
- Canada: 445
- France: 393

Concentration results:

- Top 5 countries account for 63.68% of all recorded country participation
- Top 10 countries account for 75.42%
- Country by format observations:
- The United States, India, Canada, and France were strongly movie led
- Japan and South Korea showed much stronger TV representation

Interpretation:

Netflix's catalogue appears international in scope, but not evenly global in structure. A relatively small number of countries account for most recorded country participation, suggesting that the platform's breadth is real but concentrated.



5.4 Category Concentration & Content Patterns

The fourth stage examined how content is distributed across the listed_in classification labels.

Top category labels:

- International Movies: 2,752
- Dramas: 2,427
- Comedies: 1,674
- International TV Shows: 1,351
- Documentaries: 869

Concentration results:

- Top 5 categories account for 46.95% of all category participation
- Top 10 categories account for 65.77%

Structural observations:

- The catalogue is strongly framed through both genre labels and international classification labels
- Movies and TV shows are organised differently, with TV content concentrated in a smaller set of TV specific categories

Interpretation:

Netflix's catalogue appears varied on the surface, but much of its structure is concentrated in a smaller recurring core of category labels. This suggests that catalogue breadth does not automatically mean evenly distributed diversity.

5.5 Contributor Concentration

The fifth stage explored whether Netflix's catalogue also shows concentration at the level of directors and cast members.

Most frequent recorded directors:

Rajiv Chilaka: 22

Jan Suter: 21

Most frequent recorded cast members:

Anupam Kher: 43

Shah Rukh Khan: 35

Concentration results:

Top 10 directors account for 2.26% of recorded director appearances

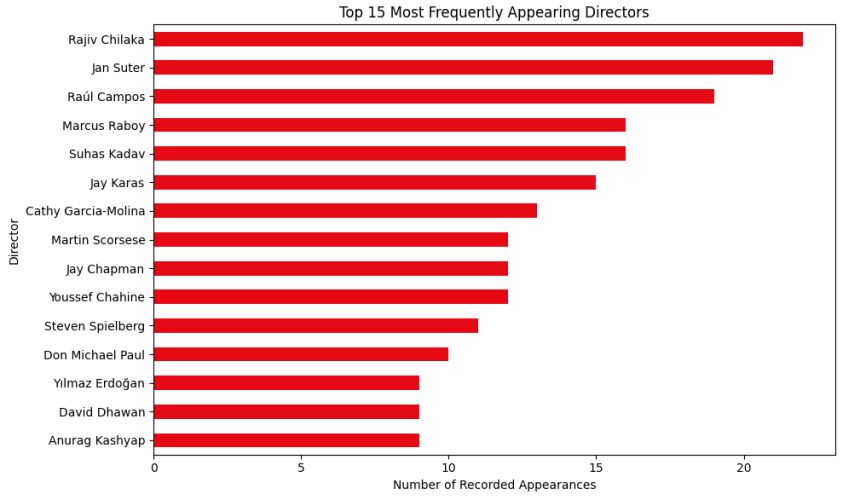
Top 20 directors account for 3.54%

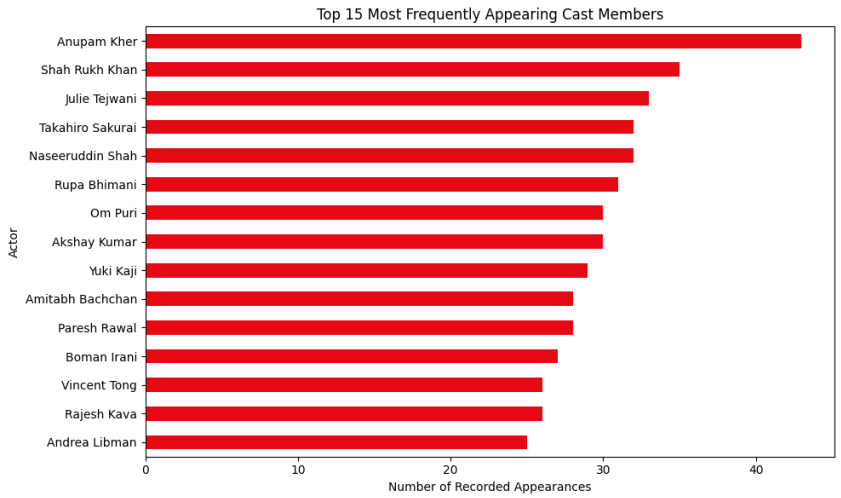
Top 10 actors account for 0.50% of recorded cast appearances

Top 20 actors account for 0.90%

Interpretation:

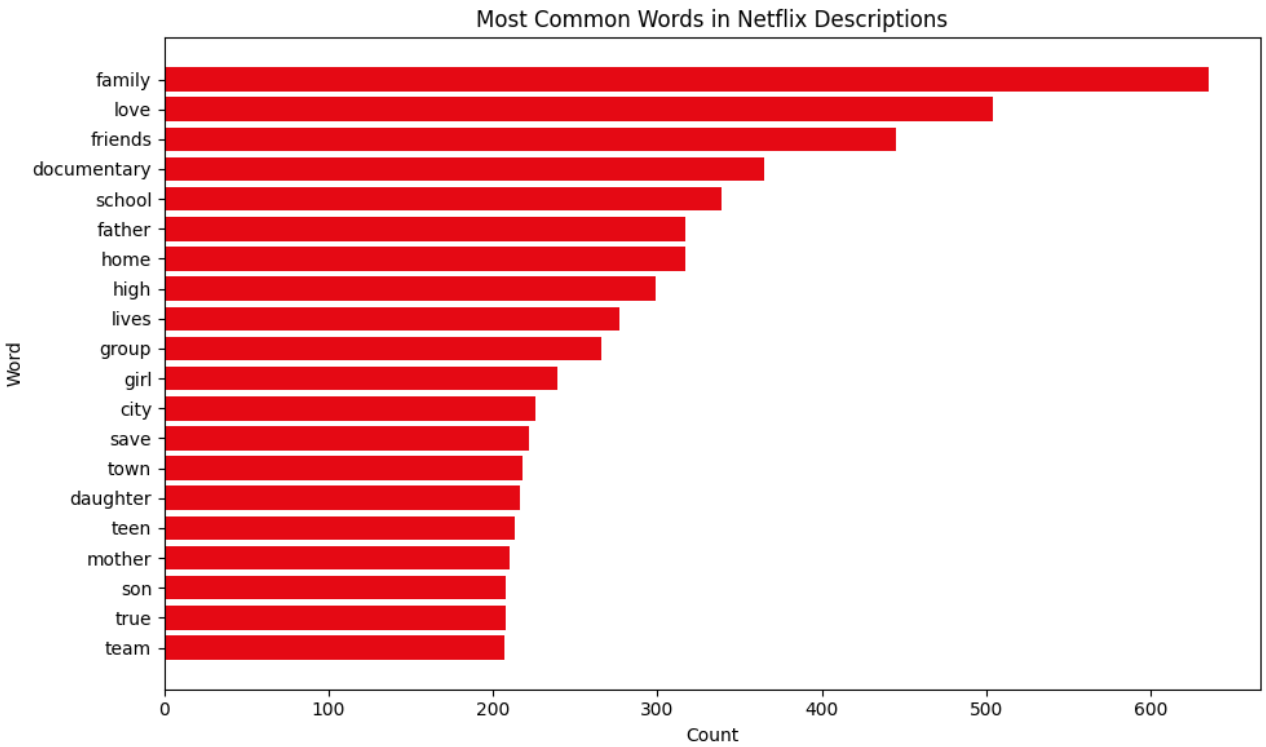
Recurring creative names are visible in the catalogue, but contributor concentration is weak overall. This suggests that Netflix's structural concentration is more strongly expressed through countries and categories than through a small set of recurring contributors.





5.6 Text Patterns and Recurring Description Themes

The final analytical section used cleaned title and description text to identify recurring language patterns.



Interpretation:

Netflix descriptions frequently foreground family, love, friendship, school, home, and documentary storytelling. This suggests that much of the catalogue is framed through human relationships, social environments, youth narratives, and emotionally recognisable settings.

This text layer reinforces the broader project finding that the catalogue may be broad in size, but repeatedly returns to a narrower set of familiar storytelling themes.

06.
Final Conclusion

This analysis suggests that Netflix's catalogue is broad in scale, but not evenly distributed in structure.

The platform expanded sharply from 2016 onward, with growth driven primarily by movies, although TV shows became a more important share of annual additions over time. Beneath this growth, the catalogue appears concentrated across both geography and classification. A relatively small group of countries accounts for most recorded country participation, while a similarly limited set of category labels accounts for much of the catalogue's structure.

At the contributor level, recurring directors and actors are visible, but overall contributor concentration remains weak. This suggests that Netflix's catalogue concentration is more strongly reflected in content origin and category mix than in repeated creative names.

The text analysis adds a final interpretive layer by showing that Netflix descriptions frequently emphasise family, love, friendship, youth, home, and documentary themes. Together, these findings suggest that Netflix's catalogue is broad on the surface, but strategically concentrated beneath it, both in structure and in the recurring themes used to frame its content.

07. Tools & Methods

Tools:

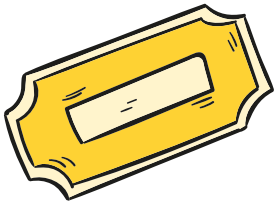
- Python
- Jupyter Notebook
- VS Code

Libraries:

- pandas
- numpy
- matplotlib
- seaborn
- collections
- re

Analytical methods:

- Data cleaning and transformation
- Exploratory data analysis
- Feature engineering
- Concentration analysis
- Categorical comparison
- Text preprocessing
- Keyword frequency analysis



08. Limitations

This project is based on metadata rather than viewership or engagement data, so it cannot explain what users watched most or what content performed best.

Other limitations include:

- missing values in director, cast, and country
- multi country titles being counted as country participation rather than exclusive totals
- category labels reflecting Netflix style classifications rather than a perfect genre taxonomy
- a small number of timing inconsistencies between release year and date added

These limitations do not remove the value of the project, but they shape how the findings should be interpreted.

09.
Next Steps

If the project were extended further, possible next steps could include:

- weighted country analysis for multi country titles
- sentiment or tone scoring for descriptions
- network analysis of actor and director relationships
- clustering of recurring text themes
- building an interactive dashboard version of the project

Contacts

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